## SOME IMPORTANT CONSTANTS IN PHYSICS

| Name | Symbols | Value |
|---|---|---|
| Speed of light in vacuum | $c$ | $2.9979 \times 10^{8}$ m $s^{-1}$ |
| Gravitational constant | $G$ | $6.67 \times 10^{-11}$ N $m^{2}$ $kg^{-2}$ |
| Acceleration due to gravity (sea level, at 45° latitude) | $g$ | 9.8 m $s^{-2}$ |
| Planck constant | $h$ | $6.626 \times 10^{-34}$ J s |
| Boltzmann constant | $k$ | $1.38 \times 10^{-23}$ J $K^{-1}$ |
| Avogadro number | $N_A$ | $6.023 \times 10^{23}$ $mol^{-1}$ |
| Universal gas constant | $R$ | 8.31 J $mol^{-1}$ $K^{-1}$ |
| Stefan – Boltmann constant | $\sigma$ | $5.67 \times 10^{-8}$ W $m^{-2}$ $K^{-4}$ |
| Wien's constant | $b$ | $2.898 \times 10^{-3}$ m K |
| Permeability of free space | $\mu_0$ | $4\pi \times 10^{-7}$ H $m^{-1}$ |
| Standard atmospheric pressure | 1 $atm$ | $1.013 \times 10^{5}$ Pa |



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